

Dr. Devesh Chandra Binwal

Halide Perovskites | Structure-Property Relationships | Magnetic Materials

New Chemistry Unit,

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Research Profile

Experimental materials chemist specializing in the design, synthesis and single-crystal growth of halide double perovskites, with particular focus on Mo^{3+} -based transition metal halide systems. Research spans dimensionality engineering (1D-3D), Dion-Jacobson and Ruddlesden-Popper 2D phases, and structure-property relationships governing magnetic ordering, optical transitions, and structural phase transitions. Experienced in SCXRD structure solution and refinement, SQUID magnetometry, and multi-technique spectroscopic characterization, with a consistent record of first-author publications in leading journals.

Key Research Contributions

- Discovery of Mo^{3+} halide double perovskites across 1D-3D for engineering magnetic superexchange and optical band gap (*Chem. Sci.* 2023, HOT Article & Pick of the Week).
- Tuning of magnetic exchange and optical absorption via halide (Cl/Br) alloying (*Chem. Mater.* 2024).
- Expansion of A-site cation in 1D halide double perovskites (*Chem. Commun.* 2025).
- Development of MoS_2 -based nanocomposites for enhanced hydrogen evolution (*ACS Appl. Energy Mater.* 2021, Cover Article)
- Ambient-stable Mo^{3+} -based Dion-Jacobson and Ruddlesden-Popper 2D perovskites (manuscript in preparation).

Publications

First-author publication: 5 | Citations: 154 | h-index: 6 (Google Scholar)

1. D. C. Binwal, P. Vishnoi, $[\text{N}(\text{CH}_3)_4]_2\text{M}^{\text{I}}\text{MoCl}_6$ ($\text{M}^{\text{I}} = \text{Na}, \text{Ag}$): Expanding the family of 1-D halide double perovskite-related materials by large A-site cation. *Chem. Commun.*, 2025, 61, 18352-18355.
2. D. C. Binwal, K. Anand, M. Sharma, K. V. Saurav, S. K. Pati, P. Vishnoi, Tailorable magnetic exchange and optical absorption in 1-D halide double perovskites $(\text{CH}_3\text{NH}_3)_2\text{NaMoCl}_{(6-x)}\text{Br}_x$, *Chem. Mater.*, 2024, 36 (16), 7719-7730.
3. D. C. Binwal, P. P. Mudoi, D. P. Panda and P. Vishnoi, Molybdenum chloride double perovskites: dimensionality control of optical and magnetic properties, *Chem. Sci.*, 2023, 14 (15), 3982-3989. (*HOT Article Collection, Pick of the Week Collection*)
4. D. C. Binwal, K. Pramoda, A. Zak, M. Kaur, P. Chithaiah and C. N. R. Rao, Nanocomposites of 1D MoS_2 with polymer-functionalized nanotubes of carbon and borocarbonitride, and their HER activity, *ACS Appl. Energy Mater.*, 2021, 4, 2339-2347. (*Cover Article*)
5. D. C. Binwal, M. Kaur, K. Pramoda and C. N. R. Rao, HER activity of nanosheets of 2D solid solutions of MoSe_2 with MoS_2 and MoTe_2 , *Bull. Mater. Sci.*, 2020, 43, 313.
6. D. C. Binwal, K. Anand, A. Bhui, K. Biwas and P. Vishnoi, Structural phase transitions, magnetic exchange and optical absorption in layered molybdenum(III) halide double perovskites (Manuscript under review).

7. K. Pramoda, **D. C. Binwal** and C. N. R. Rao, Nanocomposites of MoS₂ Nanoparticles with Carboxyl-functionalized Carbon Nanotubes and Borocarbonitrides Nanosheets, and their Electrocatalytic Hydrogen Evolution Reaction Activity, *Materials Research Bulletin*, 2022, 149, 111697.
8. P. Chithaiah, **D. C. Binwal** and C. N. R. Rao, Simple Synthesis of 2D Molybdenum Carbides Nanosheets and Their Application in Hydrogen Evolution Reaction, *Eur. J. Inorg. Chem.*, 2022, e202101086.
9. P. Vishnoi, J. L. Zuo, X. Li, **D. C. Binwal**, K. E. Wyckoff, L. Mao, L. Kautzsch, G. Wu, S. D. Wilson, M. G. Kanatzidis, R. Seshadri and A. K. Cheetham, Hybrid Layered Double Perovskite Halides of Transition Metals, *J. Am. Chem. Soc.*, 2022, 144, 6661-6666.
10. P. Chithaiah, S. Mallick, **D. C. Binwal**, A. Bhui and C. N. R. Rao, Synthesis, Characterisation, Electrocatalytic HER Activity, and Electrical Resistivity of 3R-NbS₂ Nanosheets, *New J. Chem.*, 2023, 47 (36), 17098-17102.
11. P. Chithaiah, **D. C. Binwal**, S. Mallick and C. N. R. Rao, Synthesis of Highly Crystalline Flakes of 3R-MoS₂ and 3R-WS₂ and Their Electrocatalytic and Photocatalytic Hydrogen Evolution Reactivity, *Z. Anorg. Allg. Chem.*, 2024, 650, e202400035.
12. R. Kumar, A. G. Francis, R. Stewart, P. J. Baker, R. Feyerherm, **D. C. Binwal**, B. Lake, S. K. Pati, and A. Sundaresan, Dynamic Magnetic Ground State in a Dimer Based Compound Yb₂Te₅O₁₃, *Phys. Rev. B*, 2025, 111, L140411.

Education & Research Position

Research Associate	JNCASR, Bangalore, India Aug 2025-Present Research on lead-free halide double perovskites and magnetic exchange interactions in low-dimensional systems.
Ph.D., Materials Science	JNCASR, Bangalore, India Sept 2021-Aug 2025 CGPA: 9.0/10.0 Supervisors: Prof. C.N.R. Rao, FRS and Dr. Pratap Vishnoi Thesis: <i>Investigations of HER Activity and Magnetic Exchange in Transition Metal Dichalcogenides and Halide Perovskites</i>
Visiting Ph.D. Student	University of Newcastle, Australia May-June 2025 Research under Prof. Ajayan Vinu on 2D transition-metal materials
M.S. (Engg.), Materials Science	JNCASR, Bangalore, India Aug 2019-Sept 2021 CGPA: 7.0/8.0 Supervisor: Prof. C.N.R. Rao, FRS Thesis: <i>Photochemical and Electrochemical Generation of Hydrogen Based on MoS₂ and MoSe₂, and their Adducts with Other Low Dimensional Materials</i>
M.Sc., Chemistry	Kumaun University, India 2016-2018 75.1%
B.Sc. (Hons.), Chemistry	University of Delhi, India 2013-2016 69.85%

Technical skills

Crystal Growth and Synthesis: Sealed-tube, solvothermal, hydrothermal, Schlenk line techniques, growth of single crystals of air-sensitive halide materials.

Structural Characterization: SCXRD, PXRD, XPS, SEM, TEM, UV-Vis, Raman, DSC, SQUID magnetometry.

Applications: Photocatalysis, electrocatalysis (HER), and evaluation of magnetic and optical functionalities.

Software: Olex2, APEX4, WinGX, Mercury, VESTA, GSAS-II, Origin, Fityk, MS Office.

Awards and Fellowships

1. **Sheikh Saqr Junior Fellowship**, JNCASR (2021-2023) (highly selective; 2 awards per year)
2. **DST-INSPIRE Fellowship**, Dept. of Science & Technology, Govt. of India
3. **SERB International Travel Support**, Govt. of India (2024) (**Gordon Research Conference** on Unconventional Semiconductors, USA).
4. **RAK CAM Travel Grant**, Ras Al Khaimah Center for Advanced Materials, UAE (2023) (International Workshop on Advanced Materials, UAE)
5. **Fellowship for Visiting PhD Research Program**, University of Newcastle, Australia (2025)
6. **GATE Fellowship**, MHRD (national-level competitive examination)
7. **DST-INSPIRE Scholarship**, Govt. of India (2013-2018) (top 1% of Class XII students nationally)

Conferences & Presentations (Selected)

1. **Poster: Gordon Research Conference** on Unconventional Semiconductors, USA (2024): *Molybdenum Chloride Double Perovskites: Dimensionality Control of Optical and Magnetic Properties*
2. **Poster: International Workshop on Advanced Materials (IWAM)**, UAE (2023): *Dimensionality engineered Mo(III) halide perovskites*
3. **Oral: University of Newcastle**, Australia, 2025
4. **Oral: JNCASR Seminar** (2024) - *Molybdenum-based magnetic halide double perovskites*
5. **Oral: JNCASR Seminar** (2022) - *Dimensionality Engineered Mo(III) Halide Perovskites and Their Optical and Magnetic Properties*
6. **Poster: JNCASR-Rice University Workshop** on Energy and Sustainability (2023)

Training and workshops

1. Rigaku School for Practical Crystallography (January 2022).
2. STUTI-DST Workshop on X-ray Crystallography, VIT (August 2022).
3. Workshop on Neutron Scattering and Muon Spectroscopy (Feb 2024).
4. Crash course on Next-Generation Solar Cells (March-April 2022).

Teaching and Outreach

1. **Teaching Assistant**, Recent Trends in Inorganic and Nano Materials, JNCASR (2021-2022).
2. **Peer Reviewer**, ACS Applied Energy Materials (2026)
3. **Volunteer**, National Science Day (2025), International Conference on Recent Advances in Materials (RAM-90) (2023), and JNCASR International Winter School (2023-2024).